

Vinay Vishnu Prabhu

Senior Backend Engineer & Engineering Lead | Python & PostgreSQL | Distributed Systems | Agentic & LLM Systems | AWS
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SUMMARY

Backend engineer with 5+ years building and operating production systems on Python, PostgreSQL and AWS across SaaS, IoT and EV platforms. Cut cloud infrastructure cost 85% in 3 days against a 3-week estimate; eliminated 3–4 emergency restarts a week by root-causing a file-descriptor leak; recovered 19 GB of exhausted disk by fixing runaway SQLite WAL growth; raised throughput 40% and SLA compliance from 30% to 85%. Build and run LLM agents on the Anthropic and OpenAI APIs as everyday engineering leverage — one of my own agents helped isolate the leak above. Own the full stack including React/Vue front ends, and consistently find and fix the problems teams have learned to live with, without being asked. Have led small engineering teams and run 70+ technical interviews.

SKILLS

Languages: Python, SQL, TypeScript, JavaScript, Ruby, Go, C, C#, Groovy, VBScript

Backend: FastAPI, Flask, Django, REST, GraphQL, asyncio / async concurrency, WebSockets, microservices, event-driven architecture, performance profiling

Databases: PostgreSQL, MongoDB, DynamoDB, Redis, MySQL, SQLite, query optimisation, distributed replication (LiteFS)

AI & Agentic Systems: Anthropic and OpenAI APIs, agent orchestration, tool use, structured output, prompt and context engineering, LangChain, AI-assisted debugging and code analysis

Web & Commerce: Shopify (Liquid templating, theme development, metafields, storefront customisation), React, Vue, SPA architecture, component performance profiling, bundle optimisation

Cloud & Infra: AWS (EC2, S3, SQS, DynamoDB, Lambda, IAM, CodePipeline), Azure Event Hubs, Docker, GitHub Actions, Jenkins, CI/CD, observability

Testing & Quality: Selenium, Playwright, test automation platforms, iterative delivery, incident response, code review

Leadership: Technical ownership, mentoring, sprint planning, architecture reviews, cross-team enablement

EXPERIENCE

Zeno (EV Infrastructure)

Senior Software Engineer | Oct 2025 – Present | Remote

- Eliminated 3–4 emergency production restarts per week — with zero recurrences in the months since — by root-causing a file-descriptor leak through a full audit of the code path (accelerated with an internal AI code-analysis agent I built) and shipping the fix overnight.
- Recovered 19 GB of EC2 disk that runaway SQLite WAL growth was consuming to exhaustion — starving co-located Python services and causing data loss — by correcting the checkpointing and write patterns behind it, then added a scheduled disk-audit script that emits structured diagnostics for automated triage.
- Cut monthly cloud infrastructure cost by 85% by architecting a dual-mode pull/push Partner API in FastAPI (AWS EC2, Azure Event Hubs, retry caching, lookup-table optimisation) — identified the inefficiency myself and shipped the full solution in 3 days against a 3-week estimate.
- Unblocked a weekend onboarding of 200+ customers that the team could not resolve, diagnosing and fixing the P1 defect within an hour of escalation so the rollout completed on schedule.
- Root-caused a data correctness bug across the fleet-tracking platform: event timestamps were generated at processing time rather than event origin, silently corrupting battery-swap telemetry and every downstream operational dashboard.
- Introduced the organisation's first CI/CD platform on GitHub Actions with automated rollback, taking release cycles from 10–20 minutes of manual work to zero-touch deploys, and added a Dev/QA/Prod environment tier with a build-once/promote artifact pattern so regressions are caught before production.
- Replaced LiteFS's Consul-based leasing with a custom lightweight leasing layer, reducing memory overhead and improving lease-acquisition latency across multi-region SQLite replication clusters without waiting on an upstream fix.
- Removed engineering as a bottleneck for production geodata by building a Slack-based KML ingestion tool with automated validation, diff previews and safe-write verification, letting non-technical operators ship changes independently.

Infuse (useMango AI)

Senior Software Engineer & Engineering Lead | Jul 2024 – Oct 2025 | Goa, India

- Increased throughput 40% and reduced failure rates on UMAI, a low-code test automation platform processing thousands of daily executions, by redesigning the backend around FastAPI microservices and event-driven pipelines on DynamoDB and SQS.

- Owned features end-to-end including the front end — built and shipped React and Vue components against the platform's SPA, so backend contracts and UI shipped together rather than across a handoff.
- Diagnosed a long-standing "the app is slow" complaint that had been blamed entirely on the front end. Profiling showed two independent causes — unbatched backend queries and re-rendering component trees with an oversized bundle — and fixed both, restoring responsiveness on the platform's heaviest views.
- Delivered 90% of MVP scope across 5 consecutive sprints with a 3-engineer team under aggressive timelines by introducing lean planning and tight scope control.
- Raised team velocity and delivery predictability by owning the engineering standards the team ran on — a shared Definition of Done, review expectations and lighter sprint process — while running 1:1s, performance reviews and growth plans for the engineers I led.
- Conducted 70+ technical interviews across the organisation, raising the hiring bar and shaping how the team evaluated backend candidates.
- Shipped two high-priority proofs of concept — self-driving and self-healing tests — 2 months ahead of a 3-month deadline; the rapid-prototyping approach was adopted into the core product roadmap.
- Eliminated environment drift between CI/CD and production by auditing and re-estimating 18+ epics and standardising a containerised Linux deployment stack, correcting over-optimistic architecture assumptions that were blocking reliable delivery.

Infuse (useMango Classic)

Software Engineer | Jul 2022 – Jun 2024 | Goa, India

- Reduced manual triage overhead 70% on a Python/Ruby test automation engine running 1000+ daily executions by decomposing monolithic components into containerised microservices and adding real observability.
- Raised SLA compliance from 30% to 85% in 3 months while leading a 5-engineer support engineering team, through systematic incident response improvements and platform stability work.
- Cut batch job completion time 75% (1 hour to under 15 minutes) by re-engineering legacy test execution with asyncio and multi-threaded orchestration.

Infuse

Associate Software Engineer | Feb 2021 – Jun 2022 | Goa, India

- Reduced runtime failures 40% in Windows test software handling 100+ concurrent test cases by diagnosing and resolving threading deadlocks; built a Ruby screenshotting tool 10–15% faster and ported it to a C++ DLL to cut memory footprint and latency.

PROJECTS

Jii App

Project Lead & Backend Architect — Beta on Google Play | Side project

- Built a custom payment SDK wrapping Razorpay to power the platform's subscription model — a single abstraction handling checkout, webhook reconciliation and subscription state across both Android and iOS, so product code never touched the gateway directly.
- Architected and scaled a multi-service SaaS platform (e-commerce, video reels, job listings, real-time chat) to 1000+ active beta users at 99% uptime, owning end-to-end backend design on FastAPI, AWS and MongoDB.
- Led a 10-person cross-functional team across frontend, backend and mobile.

EDUCATION

Master of Computer Applications (MCA) — Goa Business School, Goa University | 2018–2021 | GPA 7.9 / 10

B.Sc. Electronics — St. Xavier's College, Mapusa, Goa | 2015–2018 | GPA 9.0 / 10, Distinction